Home Energy Audit Report

Weatherization Assistance Program · energy audit with savings-to-investment ratio (10 CFR 440.21) · combustion safety and health & safety (ANSI/BPI-1200, state H&S plan) · ventilation (ASHRAE 62.2-2016)

SAMPLE DOCUMENT – Fictitious house and household. Test results, bills, prices and costs are illustrative values, not customer data. Calculations use a simplified hourly model calibrated to the billing history. Program submittals are produced with the DOE-approved audit tool of the grantee (e.g. NEAT/MHEA or an approved priority list).

Recommended scope: 8 energy measures · package SIR 2.41 · $1,113 per year

Energy measures incl. incidental repairs $6,951 · health & safety $1,803 · total $8,754. Predicted annual energy cost $3,039 → $1,900 (incl. new ventilation fan).

Action required

- 3 combustion safety / gas test(s) failed – must be corrected before or during weatherization and re-tested before the crew leaves each day (section 2).
- H&S cost $1,803 exceeds the per-unit threshold of $1,500: waiver from the state monitor required before the work order is released.

Client	Mrs. A. Example (fictitious) · Owner-occupied · household of 2
Address	45 Sample Street, Hamilton Township, NJ 08610 (fictitious)
Program	New Jersey Weatherization Assistance Program (DOE WAP, PY 2026) with LIHEAP weatherization funds
Agency / job no.	[Local Weatherization Agency] – DCA subgrantee · WX-2026-0147
Eligibility / priority	Income at 159 % of the HHS poverty guidelines (WAP limit 200 %); LIHEAP recipient 2025–26 · priority: Elderly occupant (age 72), High energy burden
Dwelling	Single-family detached, one-story ranch over full basement, built 1955, 1,150 ft² above grade + 1,150 ft² basement, 3 bedrooms
Climate	IECC climate zone 4A · Hamilton Township, NJ (Mercer County) · normal year 4,875 HDD65 / 1,242 CDD65
Audit date	11/04/2026 · report issued 11/12/2026
Energy auditor	[Auditor Name], [BPI Home Energy Professional – Energy Auditor, cert. no.] · [EPA RRP Certified Renovator]
Audit tool	eco-energieberater sample model (hourly, single zone, bill-calibrated) – for illustration only

3,400 CFM50 before (11.8 ACH50)	**≤ 2,300** CFM50 target (8.0 ACH50)	**2.41** Package SIR (≥ 1.0 required)	**$1,113** Savings per year (37% of energy cost)
567 therms Gas savings per year	**1,390 kWh** Electricity savings per year	**9.0% → 5.7%** Energy burden (cost / income)	**$8,754** Total job cost (ECM + H&S)

Program checks

Check	This unit	Requirement	Status
Package SIR (ECMs incl. incidental repairs)	2.41	≥ 1.0	PASS
Every installed ECM has SIR ≥ 1 in package order	AS-1 1.24, AI-1 1.92, WI-1 2.72, RJ-1 4.01, WH-1 10.71, WH-2 1.47, LT-1 11.15, RF-1 2.22	≥ 1.0	PASS
ECM + IRM cost of this unit compared with the program-year average (ACPU)	$6,951	$8,813 average	WITHIN
H&S cost of this unit (separate budget, average 14% of ACPU)	$1,803	$1,234 average · waiver above $1,500	WAIVER REQUIRED
ASHRAE 62.2-2016: continuous fan airflow ≥ required Q_fan (at target CFM50)	40 cfm	≥ 31 cfm	PASS
Calibration factor model → bills (heating)	0.89	0.70–1.30	PASS
Billing regression fit (R^2)	0.998	≥ 0.90	PASS

Limits: DOE WPN 26-1 (PY 2026 adjusted average $8,813); NJ PY 2026 H&S Plan: H&S average 14 % of ACPU, waiver for individual units above $1,500. The ACPU limit applies to the average of all units the agency completes in the program year, not to each unit. H&S costs are budgeted separately and are not part of the SIR.

1 Building and systems as found

Component	Description	Area ft²	U Btu/h·ft²·°F	ΔT factor	UA Btu/h·°F	Share
Ceiling below vented attic	R-11 fiberglass batts between 2x6 joists, gaps and compression (≈ R-9 effective)	1,150	0.080	1.0	92.0	12%
Above-grade walls, net	2x4 @ 16" o.c., empty cavities, plaster, board sheathing, clapboard	952	0.220	1.0	209.4	27%
Windows	Double-pane vinyl, 12 units	144	0.500	1.0	72.0	9%
Exterior doors	Wood panel front door with storm door, steel rear door	40	0.400	1.0	16.0	2%
Rim / band joist	2x8 rim joist, uninsulated, unsealed sill (142 ft perimeter)	107	0.300	1.0	32.1	4%
Basement wall above grade	8" concrete block, uninsulated, 1.5 ft exposed	213	0.360	1.0	76.7	10%
Basement wall below grade	8" concrete block, 5.5 ft below grade (ground-coupled)	781	0.100	0.5	39.1	5%
Basement floor slab	Concrete slab, uninsulated (ground-coupled)	1,150	0.030	0.5	17.2	2%
Air leakage	3,400 CFM50 ÷ N-factor 17 = 200 cfm natural				216.0	28%
Total					**770.5**	**100%**

Basement inside the thermal and pressure boundary. Ground-coupled components with a reduced temperature difference (factor b). Windows: Double-pane clear, vinyl frames (1990s), U ≈ 0.50, SHGC 0.55.

System	As found	Notes
Heating	Gas furnace, natural draft (Category I), 80,000 Btu/h input, installed 1996, ducts in basement	Steady-state efficiency 78.4 % measured with combustion analyzer; nameplate AFUE 78 %; distribution efficiency 0.90
Cooling	Two window AC units (bedrooms), EER 9.8	about 45% of the house cooled
Water heating	Gas storage water heater, atmospheric (natural draft), 40 gal, installed 2014, UEF 0.58, common vent with furnace	setpoint 125 °F, showerhead 2.5 gpm
Refrigerator	Top-freezer refrigerator, 18 cu ft, 1999	1,180 kWh/yr metered – 2-hour metering with logging plug meter, door closed, compressor cycles complete; annualized per state protocol
Lighting	15 lamps inventoried, 620 kWh/yr	Living room (table lamps) 3×60 W; Kitchen ceiling 2×75 W; Hall / bath 3×60 W; Basement 3×13 W; Bedrooms (already LED) 4×9 W
Thermostat	heating 69 °F, cooling 76 °F	Occupant-reported settings, no setback

SAMPLE

2 Diagnostic testing and combustion safety

Test	Result	Notes
Blower door (pre)	3,400 CFM50 · 11.8 ACH50 · volume 17,250 ft³	Single-point test, house depressurized to −50 Pa with reference to outside, basement door open, attic hatch closed, all exterior openings closed, combustion appliances off, ash/dust check. Outdoor 46 °F, Calm (< 5 mph).
Duct diagnostics	Supply registers 0.3–0.8 Pa, panned returns 1.9 Pa (leaky)	Ducts inside the basement; return leakage affects CAZ
Radon	1.6 pCi/L	Charcoal canister test, NJDEP protocol (48 h). Radon informed consent signed, EPA 'A Citizen's Guide to Radon' provided.

Combustion appliance zone – Basement (furnace, water heater, dryer)

Test	Measured	Action level	Status
Worst-case CAZ depressurization (natural-draft furnace commonly vented with water heater)	-3.8 Pa (worst -4.2 − baseline -0.4)	≥ -3.0 Pa	FAIL
Spillage – Gas furnace – natural conditions	6 s	stops ≤ 60 s	PASS
Spillage – Gas furnace – worst case	25 s	stops ≤ 60 s	PASS
Flue CO, air-free – Gas furnace	38 ppm	≤ 400 ppm	PASS
Spillage – Gas water heater – natural conditions	12 s	stops ≤ 60 s	PASS
Spillage – Gas water heater – worst case	95 s	stops ≤ 60 s	FAIL
Flue CO, air-free – Gas water heater	22 ppm	≤ 200 ppm	PASS
Oven CO, as measured – Gas range/oven	160 ppm	≤ 225 ppm	PASS
Ambient CO during inspection (maximum)	2 ppm	≤ 35 ppm	PASS
Gas leak test – Furnace gas cock union (customer side)	Combustible gas detector 1,200 ppm at union	no leak	FAIL

Worst-case set-up: Dryer on, bath/kitchen exhaust none present, air handler on, basement door closed (worst case found), house-to-CAZ pressure measured with manometer. Cause of the depressurization: Return-side duct leakage at the furnace cabinet and panned joist returns draws air out of the basement while the air handler runs. Action levels as applied in this sample per ANSI/BPI-1200-S-2017 and the NJ PY 2026 H&S Plan (oven 225 ppm as measured, ambient 35 ppm); depressurization limit per the state field guide for commonly vented natural-draft appliances.

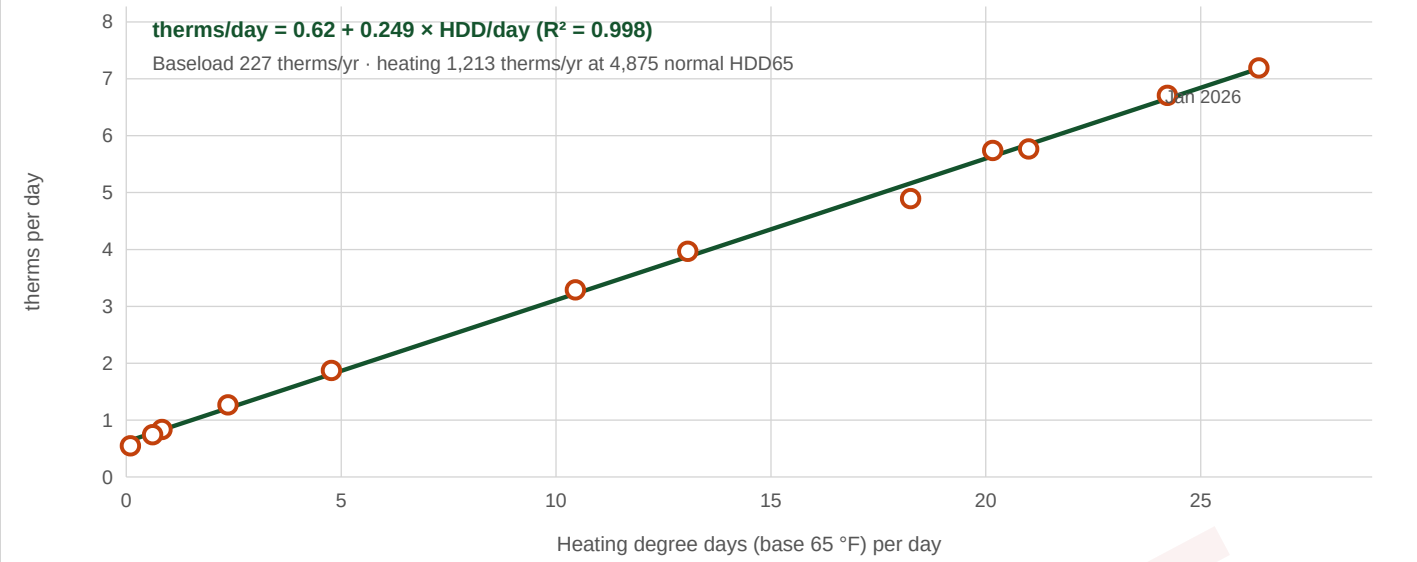
CAZ depressurization and spillage tests are repeated after the work and before the crew leaves on any day when air sealing, duct sealing or new exhaust fans could affect draft.

3 Billing analysis and model calibration

Billing period	Days	HDD65	Gas therms	therms/day	Electric kWh
Nov 2025	30	630	173	5.77	349
Dec 2025	31	751	208	6.71	371
Jan 2026	31	817	223	7.19	374
Feb 2026	28	511	137	4.89	329
Mar 2026	31	625	178	5.74	365
Apr 2026	30	392	119	3.97	336
May 2026	31	148	58	1.87	354
Jun 2026	30	25	25	0.83	414
Jul 2026	31	3	17	0.55	476
Aug 2026	31	19	23	0.74	439
Sep 2026	30	71	38	1.27	362
Oct 2026	31	324	102	3.29	346
12 months	**365**	**4,316**	**1,301**		**4,515**

Utility billing history, 12 months (fictitious); HDD65 per billing period from nearest NOAA station.

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Item	Gas, therms/yr	Notes
Billed, actual year	1,301	4,316 HDD65 – 11% warmer than normal
Baseload (water heating, cooking)	227	model: water heating 186 + cooking 30
Heating, weather-normalized	1,213	0.249 therms per HDD × 4,875 HDD65
Model heating before calibration	1,356	hourly single-zone model, normal year
Calibration factor	0.89	applied to all heating results and savings

Electric end use	kWh/yr	Basis
Refrigerator	1,180	metered
Lighting	620	lamp inventory × reported hours
Window AC	434	model, normal year
Other plug loads, furnace blower	2,282	billed total minus identified end uses
Total = billed	**4,515**	**12 months**

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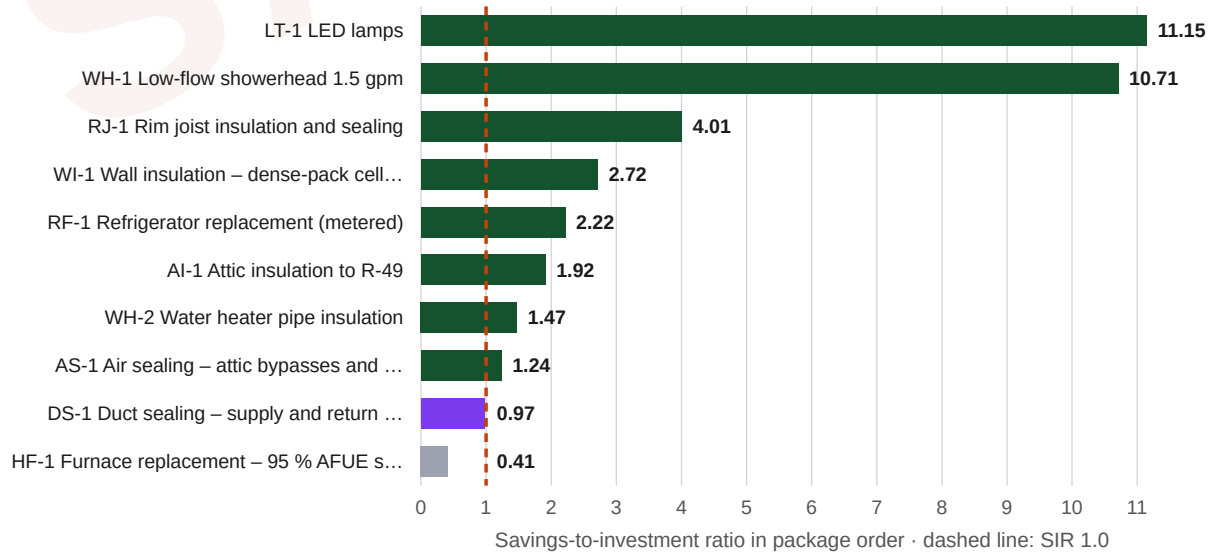
4 Energy conservation measures and SIR

Each candidate measure is first evaluated alone. The package is then built step by step: in each round the measure with the highest SIR in the current state of the house is added, so that interactions (e.g. less heating after insulation reduces the value of a new furnace) are counted. Measures that no longer reach SIR 1.0 are not installed. Incidental repairs (IRM) needed to install a measure are part of its cost.

$SIR = \Sigma_{fuel}\ (\text{annual savings} \times \text{price} \times UPV^*_{fuel,\ life}) \div \text{installed cost} \cdot UPV^* = \Sigma_{t=1..N}\ ((1+e)/(1+d))^t$ with d = 3.0% real, e = +0.5% gas / +0.0% electric

ID	Measure	Cost	Life yr	Alone $/yr	Alone SIR	Rank	Pkg therms	Pkg kWh	Pkg $/yr	Pkg SIR	Decision
LT-1	LED lamps	$88	10	$116	11.15	1	0	503	$116	**11.15**	**Install (ECM)**
WH-1	Low-flow showerhead 1.5 gpm	$48	10	$59	10.71	2	42	0	$59	**10.71**	**Install (ECM)**
RJ-1	Rim joist insulation and sealing	$449	25	$98	4.01	3	68	12	$98	**4.01**	**Install (ECM)**
WI-1	Wall insulation – dense-pack cellulose	$2,357	25	$348	2.72	4	241	43	$347	**2.72**	**Install (ECM)**
RF-1	Refrigerator replacement (metered)	$984	15	$183	2.22	5	0	795	$183	**2.22**	**Install (ECM)**
AI-1	Attic insulation to R-49	$1,676	25	$175	1.92	6	121	21	$174	**1.92**	**Install (ECM)**
WH-2	Water heater pipe insulation	$45	13	$8	1.90	7	4	0	$6	**1.47**	**Install (ECM)**
AS-1	Air sealing – attic bypasses and basement	$1,302	15	$131	1.25	8	90	16	$130	**1.24**	**Install (ECM)**
DS-1	Duct sealing – supply and return in basement	$528	15	$72	1.70	–	29	0	$41	**0.97**	**Install (H&S)**
HF-1	Furnace replacement – 95 % AFUE sealed combustion	$6,550	20	$304	0.72	–	124	0	$174	**0.41**	**Not installed**
	ECM package (installed energy measures)	**$6,951**					**567**	**1,390**	**$1,113**	**2.41**	

Install (ECM): SIR ≥ 1.0 in package order, funded as energy measure. Install (H&S): required to correct a health & safety failure, funded from the H&S budget and not part of the package SIR. Not installed: SIR below 1.0.



Prices: gas $1.40/therm, electricity $0.23/kWh (Illustrative marginal residential rates (delivery + supply) – use the client's utility tariff). 3.0 % real discount rate (DOE/FEMP, NIST IR 85-3273-40, 2025). Real energy price escalation illustrative – replace with UPV-DOE factors, residential, Middle Atlantic.

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Measure lives as applied in this sample; the state-approved audit procedure sets them. Present value of package savings $16,774.

DS-1 Duct sealing – supply and return in basement: Return-side duct leakage depressurizes the CAZ (worst case −3.8 Pa against a −3.0 Pa limit, water heater spills 95 s). Sealing is required to correct the combustion safety failure.

HF-1 Furnace replacement – 95 % AFUE sealed combustion: SIR 0.41 after the envelope measures (alone 0.72). The existing equipment tested safe; replacement is reconsidered only if it fails or becomes a health & safety issue.

5 Health and safety

ID	Item	Finding	Required action	Cost	Funding
HS-1	Gas leak at furnace gas cock union	Combustible gas detector 1,200 ppm at union, customer side of the meter	HVAC technician to remake union, leak-test all furnace connections with CGD and soap solution; occupant informed in writing	$95	DOE WAP H&S
HS-2	Dryer exhaust duct	Vinyl foil duct crushed and partly disconnected, lint in basement (fire and moisture hazard)	Install rigid metal duct, foil tape (no screws), exterior hood with damper; clean lint	$165	DOE WAP H&S
HS-3	ASHRAE 62.2 ventilation – bath exhaust fan	No bathroom exhaust, mildew on bath ceiling; local exhaust deficits kitchen and bath	Install the fan described under ventilation, duct to exterior (insulated duct in attic, roof or gable cap with damper), measure flow at both speeds	$585	DOE WAP H&S
HS-4	Smoke and CO alarms	No CO alarm; hallway smoke alarm dated 2012 (beyond manufacturer's lifetime)	Install combination smoke/CO alarm in the hallway outside bedrooms and a CO alarm in the basement; explain operation	$110	DOE WAP H&S
HS-5	Radon precautions	Open sump pit in basement; radon test 1.6 pCi/L (below 4.0 action level)	Airtight sump pit cover with sealed penetrations, seal slab cracks and floor penetrations	$140	DOE WAP H&S
HS-6	Lead-safe work practices	House built before 1978, painted clapboard and plaster (lead presumed)	EPA RRP lead-safe practices for drilling siding and any interior disturbance: containment, HEPA cleanup, cleaning verification, records	$180	DOE WAP H&S
DS-1	Duct sealing – supply and return in basement	Return-side duct leakage depressurizes the CAZ (worst case −3.8 Pa against a −3.0 Pa limit, water heater spills 95 s). Sealing is required to correct the combustion safety failure.	Seal return plenum, filter slot (cover), cabinet seams and panned joist returns with mastic and fabric; seal supply plenum and takeoffs. Re-test CAZ worst case with air handler on.	$528	DOE WAP H&S
	Total health & safety			**$1,803**	

Deferral screening

Condition	Finding
Roof leaks / structural defects	None observed
Mold / moisture	Mildew < 10 ft² on bath ceiling – cleaned, addressed by HS-3
Asbestos / vermiculite	No vermiculite in attic; duct tape is foil (no suspect ACM)
Knob-and-tube wiring	None – NM cable in attic
Sanitation / pests	None
Occupant cooperation / informed consent	Signed; occupant agrees to ventilation

Result: no condition requires deferral. A condition found later (hidden structural damage, asbestos, knob-and-tube wiring) stops the work until the auditor has reassessed the job.

Ventilation – ASHRAE 62.2-2016 (existing building)

Step	As found	After weatherization
Floor area / bedrooms	1,150 ft² / 3	same
Q_tot = 0.03 · A + 7.5 · (N_br + 1)	64.5 cfm	64.5 cfm
Blower door Q50	3,400 CFM50 (test)	2,300 CFM50 (target)
Q_inf = 0.052 · Q50 · wsf (0.53) · (H/8.2)^0.4, H = 9.5 ft	99.4 cfm	67.2 cfm
Local exhaust Kitchen: required 100 cfm, window credit 20 cfm	existing 0 cfm → deficit 80	proposed 0 cfm → deficit 80
Local exhaust Bathroom: required 50 cfm, window credit 20 cfm	existing 0 cfm → deficit 30	proposed 80 cfm → deficit 0
Q_tot,adj = Q_tot + 0.25 · Σ deficits	92.0 cfm	84.5 cfm
Φ (unbalanced: Q_inf / Q_tot,adj)	1.08	0.80
Required fan airflow Q_fan = Q_tot,adj − Φ · Q_inf	**0 cfm**	**31 cfm**

Step	As found	After weatherization
Installed continuous fan	none	**40 cfm PASS**

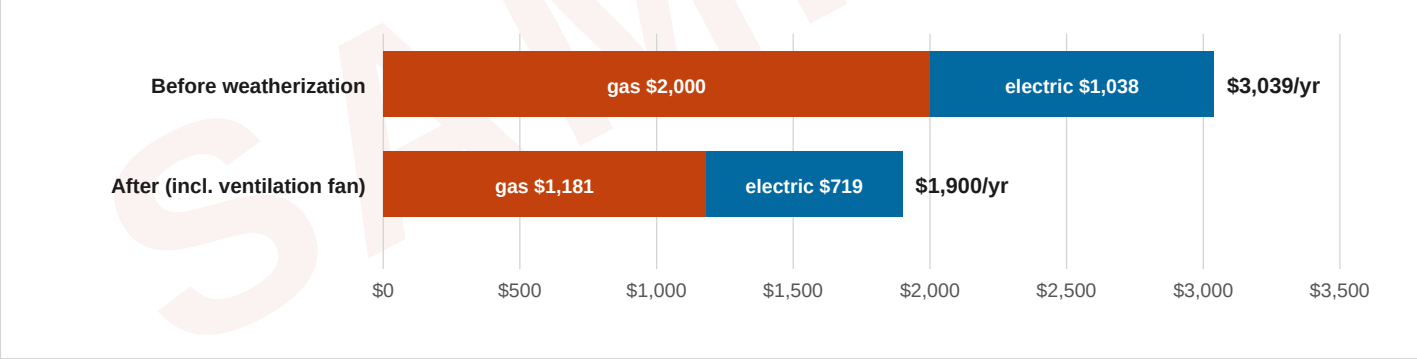
Fan: ENERGY STAR bath fan, 80 cfm boost on occupancy/humidity sensor, continuous low speed 40 cfm, ducted to exterior with backdraft damper. ASHRAE 62.2-2016 with Normative Appendix A (existing buildings); weather and shielding factor wsf illustrative – NJ requires the RED ASHRAE 62.2-2016 calculator with target and final blower door readings. The calculation is updated with the final blower door result; if the house ends up tighter than the target, the fan rate is increased. ASHRAE 62.2 does not account for unusual pollutant sources and does not guarantee indoor air quality.

6 Cost summary and predicted results

Budget category	Cost	Program reference
Energy conservation measures (incl. incidental repairs $359)	$6,951	counts toward ACPU average $8,813
Health & safety (separate budget)	$1,803	average limit $1,234 · unit waiver above $1,500
Total job cost	**$8,754**	

Program support, audit and inspection costs of the agency are not included.

Normal-year energy use	As found	ECM package	After all work (incl. H&S, ventilation)
Space heating, therms	1,213	693	674
Water heating, therms	186	139	139
Cooking, therms	30	30	30
Window AC, kWh	434	341	343
Lighting, kWh	620	117	117
Refrigerator, kWh	1,180	385	385
Other electric, kWh	2,282	2,282	2,282
Total gas, therms	**1,429**	**862**	**843**
Total electricity, kWh	**4,515**	**3,125**	**3,127**
Site energy, MMBtu	**158.3**	**96.9**	**95.0**
Annual energy cost	**$3,039**	**$1,926**	**$1,900**



The last column includes the H&S duct sealing and the continuous 40 cfm exhaust fan (141 cfm combined air exchange). Predicted savings are engineering estimates; program evaluations often find realized savings below audit predictions. Calibration to the bills reduces, but does not remove, this gap.

7 Method and limitations

Hourly heat balance of one thermal zone (basement included) over a synthetic normal weather year (Synthetic normal year calibrated to roughly 4,900 HDD65 (central New Jersey) – an approved audit tool uses TMY weather for the nearest station). Heating and cooling loads from UA, natural infiltration (CFM50 ÷ N-factor, combined in quadrature with exhaust fan airflow), internal gains and solar gains; no thermal mass. Heating results scaled to the weather-normalized billing history (fixed 65 °F base). Water heating from hot-water use and UEF; baseload measures from metering and inventories. Utility income-qualified programs use their own measure lists and cost tests; the SIR screen shown here follows 10 CFR 440 for the DOE program.

8 Client education and consent

Item	Done
Audit results, recommended measures and measures not recommended explained	☐
Combustion safety findings and CO risks explained; CO and smoke alarm operation shown	☐
Radon informed consent signed; EPA 'A Citizen's Guide to Radon' provided	☐
Lead-safe work practices and pre-renovation information ('Renovate Right') provided	☐
Ventilation fan: purpose, controls, service switch and cleaning explained; occupant agrees not to switch it off	☐
Thermostat settings and water heater temperature discussed	☐
Schedule, access and occupant protection during work agreed	☐

Energy auditor: [Auditor Name]
signature / date

Client: Mrs. A. Example (fictitious)
signature / date

Agency approval (work order release)
signature / date

SAMPLE